Howie Wang

Solutions engineering & implementation — I connect what a business needs with the systems that deliver it.

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PROFILE

Final-semester Master of Information Technology student at UNSW (graduating December 2026) moving into solutions engineering and implementation. Alongside study, I have spent the past year as the single point of contact between a biotech company's owner and the platform that runs the business: understanding the need, choosing and integrating eleven partner systems — payments, ERP, cloud and the WeChat ecosystem — getting everything live (launched 10 October 2026), training staff and supporting it through 67 releases. I can read an API document and a contract, explain either to a non-technical owner, and prototype when a slide isn't enough. Looking for a junior solutions engineer, implementation consultant or technical consultant role.

WHAT I BRING

Integration & implementation	Integrated 11 partner systems — WeChat Pay v3 (JSAPI / H5), Alipay (PC / JSAPI), AdaPay, WeChat Login & Order Centre, WeCom SSO, Aliyun SMS / OSS / MTS, Hupun ERP and a store platform; configured environments, domains, HTTPS and CDN; implemented request signing (HMAC-SHA1 / SHA-256, MD5) where no SDK existed
Discovery & requirements	Translated an owner's goals into 17 scoped product areas — storefront, loyalty programme, dealer network, commission scheme, anti-counterfeit verification, content distribution, finance reporting — with a phased plan
Go-live & support	Ran the production launch against a readiness checklist; planned a zero-downtime data migration around dealers' business hours; health checks, runbooks and audit logs; first line of support for the client's staff through 67 releases
Enablement & training	Wrote an illustrated operations guide that maps every Mini Program screen to where staff change it, four role-based staff manuals, plus 17 operational documents; ran structured feedback rounds and turned them into prioritised releases
Data & reporting	Finance reconciliation and downloadable reports for the accountant; member and customer exports; a map dashboard with country → province → city → district drill-down
Compliance & platform administration	WeChat platform requirements (real-name verification, phone-login compliance), Mini Program entity-change filing, pre-packaged food seller registration paperwork
Communication	Bilingual English / Mandarin; worked remotely from Australia with a China-based owner, staff, accountant and vendors' engineers — each got the document they needed

SELECTED INTEGRATIONS

PARTNER	WHAT IT DOES FOR THE BUSINESS	NOTABLE
WeChat Pay v3 · Alipay · AdaPay	Consumer and dealer payments; refunds through the original channel	One payment abstraction across three providers; payments arriving after a cancellation are refunded automatically; public-key verification of WeChat Pay notifications
WeChat Login · Order Centre	One-tap sign-in with phone number; orders visible in WeChat's native "My Orders"	Detects new vs. existing customers and merges accounts
WeCom (Enterprise WeChat)	Staff single sign-on to the console; contact directory sync	OAuth callback flow

PARTNER	WHAT IT DOES FOR THE BUSINESS	NOTABLE
Hupun ERP · store platform	Paid orders pushed to ERP and store systems for fulfilment	MD5 / HMAC-SHA256 signing implemented by hand; retryable outbound queue so a partner outage never blocks a sale
Alibaba Cloud SMS · OSS · MTS	Notifications, media storage with direct upload, video transcoding	HMAC-SHA1 signing without the vendor SDK; resumable, accelerated uploads (~21× faster from overseas); videos on a custom domain

EXPERIENCE

YuanQiPai (元气湃) — Sole technical lead: discovery, implementation, go-live and support

Oct 2025 – Present · Contract | Biotech company selling health products through a dealer network and directly to consumers on WeChat

- **The problem.** The business ran on disconnected tools: no single view of members, dealers, orders or commissions, and every marketing change needed outside help.
- **What I implemented.** One platform, four products — a WeChat Mini Program storefront (60 screens), a staff console (45 pages, 100+ screens, four staff roles with two-person approval for money), the API behind them and a stand-alone customer registration tool — covering membership tiers, coupons, points wallet, dealer ordering and inventory, commission settlement, anti-counterfeit verification, marketing assets, training content, complaints and finance reporting.
- **Integration.** Connected 11 partner systems for payments, sign-in, ERP fulfilment, cloud media and notifications; handled the hard parts — callbacks that arrive late, retries, idempotency, refunds — so the business never sees them.
- **Go-live.** WeChat platform compliance and filings, domain / HTTPS / CDN, a launch-readiness checklist, and a production data migration with zero downtime.
- **Enablement & support.** Wrote the operations guide and four role-based staff manuals, ran structured feedback rounds with the client and shipped fixes across 67 releases; on call when something broke.
- **Running it.** After launch: an end-of-life platform upgrade with ~30 seconds of downtime, daily tested backups, a security review with credential rotation, and video moved off the app server.
- **Decisions that mattered.** A drag-and-drop page builder so marketing never waits for a developer · commission rules as console settings, not code · automated refunds for late payments instead of manual reconciliation.
- **Outcome.** Live since 10 October 2026 with 900+ members (1,400+ users) and 900+ customer registrations; the client's staff make day-to-day changes themselves.

Customer Registration System — Sole technical lead

May – Jul 2026 · Contract | Stand-alone tool replacing paper customer forms for offline sales across 20 countries

- Designed a mobile form staff can hand to a customer: country-aware fields, a 3,423-district address selector, ID photo capture and a product-feedback survey.
- Gave the owner a live dashboard — a China map with three-level drill-down, time filters, search and CSV export — instead of a spreadsheet.
- Integrated it with the main platform through a signed webhook with retry, matching customers on ID document then phone so no record is ever duplicated; deployed independently on Alibaba Cloud with HTTPS.

TECHNICAL FOUNDATION

Node.js 22 · Express · PostgreSQL 18 · SQLite · REST APIs · OAuth · Webhooks · WeChat Mini Program · HTML / CSS / JavaScript · Alibaba Cloud (ECS, OSS, SMS, MTS) · Docker · Nginx · DNS / TLS · GitHub Actions · Jest (760+ automated tests)

EDUCATION

Master of Information Technology — University of New South Wales (UNSW Sydney) · expected completion December 2026

Referees available on request.